

Consumer devices for eCommerce access

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It is widely predicted that electronic commerce will have a far-reaching impact on businesses in the next few years. For this to occur it will be essential that users are able to access and use eCommerce services when and where they wish, at times and places that are convenient to them. It is likely that increasingly many of the access devices used will not be standard desktop PCs. In this paper the range of potential devices that could in future be used to access eCommerce services is considered, and some of the issues in supporting this diverse range of devices are examined.

1. Introduction

During the 1990s the personal computer, mainly in the form of the Microsoft/Intel 'Wintel' machine, has been the dominant means of access to computing and information services. The use of Internet services, particularly those based on World Wide Web (Web) technologies, has increased rapidly in recent years, and PCs have been the predominant access device. However, it is now being increasingly accepted that as use of the Internet spreads throughout the population, many will find their needs are better met by specialised 'information appliances' as access devices [1]. The market research company IDC, in a report entitled 'The End of the PC-centric Era', has predicted that by 2002 the unit sales of such appliances will start to exceed those of PCs.

These information appliances will offer the convenience, ease of use and reliability that PCs have failed to provide. They will take a variety of forms and will be user-centric, performing a specific function required by the user conveniently and simply. Initially they will include versions of 'smart' screenphones, mobile phones, personal digital assistants (PDAs), hand-held (palm) computers, television set-top boxes, game consoles, and kiosks. In time it is expected that an increasing diversity of devices will appear, including in-car systems, wearable devices and enhanced capabilities incorporated into existing appliances, e.g. the Internet-enabled microwave oven.

In addition to devices for providing people with access to Internet services, other appliances will be Internet enabled. For example, a central heating controller would be able to access weather forecast data to enable it to predict future heating requirements, and would itself be remotely accessible to enable the homeowner to adjust temperature

settings. These developments lead to a model of computing becoming known as pervasive computing [2], which for the user will mean being able to gain immediate access to information and services, anywhere and at any time.

Electronic commerce (eCommerce) services, supported within companies by general eBusiness applications, will be a major driver for the expansion of the use of the Internet. In the early stages most eCommerce revenues will be generated from business-to-business transactions. One such emerging sector is application service providers (ASPs), who deliver packaged business software for on-line rental and use [3].

The ability of a company to deliver its services to a diverse range of access devices, anywhere, at any time, will be a key aspect of competition in eCommerce. This paper describes and compares the types of device becoming available and considers the technology that will be required to support their use, including the use of new reflective software techniques to produce self-adaptive services, and contains:

- overviews of current eCommerce access devices, including a review of both fixed and mobile forms,
- techniques for delivering eCommerce services to differing access devices, considering emerging architectures and some adaptation technologies,
- a short discussion of some future considerations for eCommerce access devices.

2. Overview of access devices

2.1 General requirements

The fundamental requirement of an access device for eCommerce services is to provide a highly usable interface, particularly as the availability of such services spreads beyond the computer-literate users. A Web browser running on a PC provides the established form of interface, but such interfaces increasingly assume a high-resolution full-colour screen. When the same service is accessed on a device with more limited capabilities, e.g. low-resolution monochrome screen, it becomes very difficult to use. The Web, using the hypertext mark-up language (HTML), was designed to be accessible by a range of devices, including text-only browsers, e.g. Braille machines for blind users. However, this aspect has been subverted on many Web sites by the quest for visually appealing and distinctive interfaces. The proliferation of diverse access devices will require a radical rethink in this area.

Another requirement for access devices is support for a communications protocol to communicate with the service provider. For Internet access this will be TCP/IP or other IP-based protocols, but for other forms of access, particularly mobile, more specialised protocols such as SMS will be used.

Also, eCommerce services place high demands on security. Aspects relevant to user devices include access security, which may be provided via a personal authorisation device (e.g. smart card), and secure data transfer, that requires the device to support suitable protocols (e.g. the secure sockets layer (SSL) within TCP/IP). These security issues will not be considered in detail in this paper.

The increasing range of possible access devices makes detailed consideration of all options difficult. To provide a focus for this section, the following guidelines have been used.

- What existing access device types are currently popular with potential eCommerce users, e.g. what type of devices could be used to receive electronic bank statements?
- What new access device types are emerging from the industry, e.g. new form factor mobile devices?
- What new access technologies are being developed within the industry, e.g. new communications infrastructures?

Access devices will be considered in two categories, fixed and mobile (portable) devices. Mobile devices are likely to make use of wireless communications links.

2.2 Fixed

Most current eCommerce services are designed to be accessed using the capabilities of PCs equipped with the latest Web browsers, largely in the form of fixed desktop machines with dial-up (PSTN or ISDN) network access. This form of eCommerce access is unlikely to address the mass-consumer market, and a variety of new forms of device are appearing.

Internet screenphones

Internet screenphone (or Web phone) products have been announced by various vendors, including Matra Nortel [4], Alcatel [5] and Samsung [6], and will become available in 1999. They combine the facilities of a telephone with an Internet access device, by providing a handset and keypad, an LCD touch screen (typically VGA resolution), a small profile keyboard for data entry, and a smart-card slot for secure access. They contain a processor plus RAM, with ROM for basic software and flash RAM for updatable software and configuration data. Various options for Internet connection are likely to be offered, including PSTN via modem, ISDN, and, in the longer term, asymmetric digital subscriber line (ADSL). Figure 1 shows a BT-designed prototype first demonstrated in 1997, and a recently released commercial device, the Matra Nortel Tel@phone®.



Fig 1 BT prototype screenphone (1997) and Matra Nortel Tel@phone.

The devices will provide advanced telephony facilities, including integral directories with direct dialling, incoming and outgoing call logs, and user-friendly screen-based interfaces to network services such as call waiting, call diversion and three-way calling. In addition they will provide full Internet access, including the Web and the potential for convenient unified messaging, combining access to voice, fax and e-mail messages.

These devices are clearly aimed at the consumer and small business markets, but they also have the potential in specific areas of corporate markets to combine the functions of the existing desktop PC and telephone.

The Internet Screenphone Reference Forum (ISRF) [7], whose members are device manufacturers, telecommunications operators and software companies, is developing reference specifications for Internet screenphones. These cover requirements for the devices, applications, core services supported and service management aspects. One specific issue being addressed is event notification, whereby a service can notify the user of the arrival of a message, e.g. e-mail. There is no standard way of providing this capability over a PSTN network, but BT could use the no-ring call facility, possibly combined with CLI. For an ISDN network, use could be made of the proposed AO/DI (always on/dynamic ISDN) features [8], where the D-channel maintains a permanent low-bandwidth connection, only bringing in one or both 64-Kbit/s channels when required. AO/DI would enable e-mail and Web-push data to be delivered, along with the notification.

Set-top boxes

The set-top box (STB) is a peripheral device that connects to a standard TV set and transcodes incoming signals for display on the TV. STBs are used to connect to cable and satellite services, and more recently to digital TV services via satellite or terrestrial signals. For interactive services the connection from the user to the service is typically provided via a modem in the STB connected to a telephone line. ADSL running on standard telephone lines has also been used to deliver high bandwidth interactive services via an STB.

These types of two-way STBs provide access to video and multimedia services, and eCommerce services such as home shopping and home banking. The user input device can be a standard remote control, or possibly a keyboard connected by an infra-red link to the STB. User access security can be provided by smart cards, which could also be used for eCommerce applications, such as loading cash cards within home-banking applications.

STBs were first introduced in the 1970s with the emergence of cable TV companies in the US, providing numerous TV channels. The cable TV providers defined the early STB specifications, but these were mainly proprietary rather than agreed standards. However, there has been some government standardisation efforts in the USA [9].

A recent innovation is an STB, also often termed net-top box, which provides access to the Internet on the TV. Using a subscription-based service, a proprietary browser is provided, which enables Web pages to be browsed in a similar way to a PC but with some adaptation of the text and images for the limited capabilities of the TV display. E-mail and other applications can also be used, but as there is no local storage in the TV, data has to be stored with the service provider or in the STB.

The battle for dominance of the STB vendor market is being strongly influenced by the choice of operating system (OS). Web TV [10], now owned by Microsoft, uses Windows CE, whereas other vendors such as TCI [11] and Hitachi are looking to use the PersonalJava OS. Sun, the developers of Java technology, have now released the Java TV API to encourage further Java developments in the STB market. Internet TV services are available in parts of the USA, and are likely to appear in the UK.

High street devices (automatic teller machines and kiosks)

Automatic teller machines (ATMs), as shown in Fig 2, became widespread during the early 1980s, providing some bank services (e.g. cash dispensing) on a 24-hour basis.

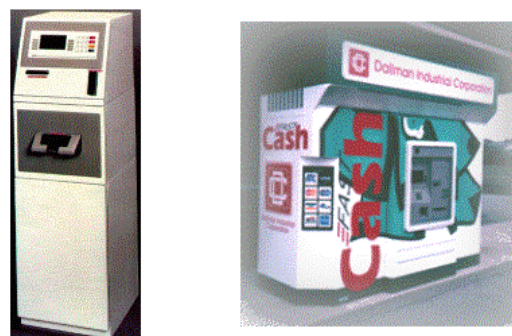


Fig 2 Stand-alone and mounted ATMs.

ATMs have recently seen another increase in numbers due to technology advances, with devices becoming cheaper and easier to accommodate. By placing them in supermarkets and other locations, they have been more effective. In the US, between 1993 and 1997, the number of ATMs rose by 66% [12]. Most high streets ATMs still offer similar banking services to those provided when they were first introduced, but future developments will be influenced by the evolution of electronic cash handling.

ATMs have been largely designed for ruggedised high street usage. Sun have announced that its PersonalJava OS can support mini ATM devices for domestic use [13], where the ATM enables secure and convenient access to home banking services, including electronic cash withdrawals and transfers between personal smart cards. High street ATMs of the future will also need to be able to manage e-cash cards.

BT has been involved with the trials for e-cash with Mondex [14] where smart cards could be charged up over the telephone and used in selected retail outlets.

Multimedia public kiosks offering Internet access in the high street are relatively new devices. They can offer users convenient access to eCommerce services while on the

move, without the need to carry personal devices. They are strategically located in places where users are likely to require such services, for example, at airports for car hire, hotels for booking travel and theatre tickets, and shopping centres for banking services. Summit Research Associates [15] have defined two types of network-connected kiosks (referred to as Internet kiosks):

- Internet-based kiosks — these are kiosks that offer specific Web information and services via a commercial wrapper, hiding the complexities of traditional PC Web browsers,
- Web payphones provide pay-as-you-go Internet access, for use of e-mail and the Web — also known as public Internet terminals or Internet payphones, they could replace existing payphones in suitable locations.

Users of kiosks get up-to-date Web information and can be put in contact immediately with any of the advertisers' sales desks, and make payments via credit card. Printouts can also be obtained on some devices. The service is usually free to the user, funded by advertisers who are being provided with new delivery channels. Many emerging kiosks are based on Microsoft Windows CE technology but other vendors offer a Java equivalent platform [16, 17], examples of which are shown in Fig 3.



Fig 3 BT Touchpoint kiosk and two Java-based kiosks.

BT launched a public multimedia kiosk service called Touchpoint in 1997 [18] (Fig 3). It offers high street shopping, entertainment and information services. Market research showed that 400,000 people use the kiosk every month, 96% of respondents found the service easy to use, and 43% were repeat users. BT is looking to increase its kiosk base with the introduction of the BT Multimedia Payphone.

Network computers

The network computer (NC) [19] is a device that connects to the same peripherals as a PC, i.e. large display, full-size keyboard, and mouse, but has a radically different approach to the storage and management of applications and data. Based on Sun's Java technology, the NC boots up by connecting to a network server, downloading or updating

the operating system software and then downloading all application programs required. This ensures that the latest software versions are always used. All user data is also stored and managed at the server, providing high levels of reliability and security. NCs have begun to be used within some commercial enterprises, but devices for use in the consumer and small business markets are likely to provide the same capabilities in different physical forms, including the Internet screenphone.

Fixed-line telephones

These ubiquitous devices must be considered as the largest available eCommerce access device currently available. Telephone device market density, called teledensity, is the number of main telephones accessible by 100 people. On a global scale, it has been estimated that over half the world's populations are living in countries with teledensity values less than one. In contrast, highly developed countries have a teledensity value of 99.

Telephone-based technologies, such as interactive voice response (IVR) or keypad driven, have been used in attempts to reach significant market segments with eCommerce services such as information retrieval or ticket purchasing, but in general they have failed. Market research shows that people dislike complex IVR menus, and many transactions are never completed. The potential of information appliances is to deliver these services in a much more user-friendly way.

2.3 Mobile

Many of the recent changes in access device technology provide greatly increased support for user mobility. Mougayar [20] predicts that wireless devices like mobile phones will become the next point of sale (POS) devices for eCommerce, as the use of mobile phones spreads throughout the population.

Other industry indicators also support this trend towards mobility. For example, Semico Research Corp [21] estimates that handheld computing will be a \$7.5 billion market by 2003. Mobile computing devices (MCD), such as laptop, notebook and handheld PCs, palmtop computers, PDAs, and smart mobile phones, pagers and other messaging devices, will provide alternative forms of access to eCommerce services.

New technologies, such as touchscreens, handwriting recognition, voice recognition, graphical controls, and audio and video output are providing the possibility of radically different human/computer interfaces on MCDs. The ease of use of device interfaces is likely to be the key issue for their acceptance by consumers.

A variety of new physical forms, tailored to the user's own environment, have been produced. Figs 4 and 5 show three devices that could be classed as PDAs, but have very different forms. The Psion 5 has a conventional keyboard and the 3Com Palm III uses a touch screen with character recognition. The SmartQuill, developed by BT, is a pen-based device that captures text data directly from handwriting, and has an integral LCD to display text [22].



Fig 4 3Com Palm III and Psion 5.



Fig 5 BT SmartQuill.

An issue for current users of MCDs is the complexity of connecting to networks, both fixed and mobile. Larger devices, such as laptop PCs, can use standard PCMCIA modem cards, but for smaller devices, such as palmtop computers, it can be more complicated. For example, to connect a Palm III to a GSM network could involve a 'Snap-On' modem, an interface adapter, a GSM phone and connecting cables.

Infra-red connection between devices and mobile phones, using the infra-red data association (IrDA) protocol, can simplify this complexity, but requires line-of-sight association between devices. An initiative called Bluetooth [23], launched in May 1998, aims to produce short-range radio-link connections between devices. It will utilise unrestricted 2.45 GHz ISM 'free-band' radio frequencies, and provide for rapid *ad hoc* automatic connections between devices. The Bluetooth Forum aims to produce its first specification in the spring of 1999.

Another possible approach is to build the PDA functionality into the mobile phone, e.g. the Nokia 9110 communicator (Fig 6). This provides e-mail, fax, SMS, Web and PDA capabilities in a 'pocket-size' device. The need to have a reasonable size screen and keyboard

constrains the minimum practical size for such a device, but further innovations are likely to address these issues.

Wrist watches with built-in PDA and communications facilities have begun to appear. Some commercially available watches have significant computing capabilities, for example Seiko's Ruputer (Fig 7) has a 16-bit processor, 128 Kbytes of RAM and 512 Kbytes of ROM, plus a 38-kbit/s infra-red port. Some Japanese watch manufacturers have built in communications capabilities for their own Personal HandyPhone System (PHS), making the devices into mobile phones. For Europe, Swatch is currently looking to build GSM capability into their watches.



Fig 6 Nokia 9110 communicator.



Fig 7 Seiko Ruputer computer watch.

Much research is going on into other forms of wearable interface devices and the capabilities they would provide. Some examples are shown in Fig 8, including monocular and head-mounted displays and a glove-based input device. While these are currently seen as exotic devices, they may form the basis of practical future devices [24].

Communications

The requirement of fast data access for mobile users has spurred technology advances in wireless networks. GSM currently provides up to 9.6 kbit/s, but the next generation will increase this to 64 kbit/s. The General Packet Radio Service (GPRS) [25] will enable IP services to be delivered over a GSM-type packet-switched network.



Fig 8 Wearable interface devices.

Future mobile networks, such as Universal Mobile Telecommunication System (UMTS) [26], aim to increase data rates to 1—2 Mbit/s. UMTS may become a broadly used network, offering both higher bandwidth and additional services, such as:

- terminal mobility, by which it should be possible for devices to access services from different locations while in motion, with the network locating and identifying the device and user,
- personal mobility, enabling a user to access services at any fixed or mobile terminal on the basis of a personal telecommunications identifier,
- service mobility, providing access to the same set of services from any connection point within public and private environments (this concept is also known as the virtual home environment (VHE)).

The performance of mobile networks is likely to be a key factor in determining the take-up of mobile access to eCommerce services. However, the diversity of network standards still remains one of the biggest problems for mobile device manufacturers. For example, the PalmOS pdQ smartphone from Qualcomm is the first integrated palm computer and mobile phone [27]. It is due to be released in mid 1999, but it will have limited European use because it only works with CDMA networks. The standardisation of UMTS is being delayed by disagreements between Europe, the USA and Japan, and this is likely to delay its implementation.

Operating Systems

The current view [28] is that no single operating system (OS) will dominate the mobile computer market. There are four main contenders.

The Palm OS, owned by 3Com [29], is the basis for the successful Palm palmtop device range. At the end of 1998 it had the largest PDA user base, with 41.4% of the world-wide hand-held market [30] and 70% in the US [31].

Microsoft released Windows CE [32] a few years ago with the vision that it would suit a range of small devices, including hand-held PCs, palmtop computers and auto PCs (e.g. car stereos). It has failed to make the market impact expected in 1998, but with many established hardware vendors (including Philips, Casio, Hewlett Packard and Sharp) increasing their device range, and enhanced OS releases, it may see significant market growth in 1999.

The third vendor is Symbian with its EPOC operating system. The Symbian alliance [33] includes Psion, Motorola, Ericsson, Matsushita and Nokia, producing an interesting combination of PDA, CPU and mobile phone technology manufacturers. EPOC is used in the current Psion PDAs, but no mobile devices using EPOC have been announced yet.

Java OS is the fourth option, and Sun have released a specification detailing a Java Application Environment, PersonalJava [34], which is aimed at consumer devices. Its architecture is more suited to constrained memory and processing environments, and can optionally include parts of the full Java application programming interface (API). New devices running PersonalJava operating systems are only just appearing, and it is difficult to predict their market take-up. Sun are also producing other Java APIs for speech, sound, telephony and multimedia. These are likely to migrate into mobile devices.

The other consideration with Java is that Java Virtual Machines (JVM) may be available on top of existing OSs. Symbian have released details of their intended implementation of a PersonalJava JVM, to run on the EPOC OS [35] and a JVM for Windows CE OS is available.

WEB access

Most existing eCommerce services use Web access, and, as this is likely to continue, mobile computer Web browsing capabilities will be important.

Laptop and notebook computers use standard Windows PC Web browsers, although performance over a slow wireless network can be frustrating for pages with high graphics content. Web browsers for smaller devices, such as PDAs, are quite varied and are tightly linked to the device platform. The four main PDA OS vendors are compared below:

- 3Com relies on independent software vendors to supply Web-browsing capabilities for their Palm devices, including Smartcode's HandWeb [36], which

offers a simple text-only interface, and ProxiNet's ProxiWeb browser [37], which performs some proxy image conversion,

- Microsoft Windows CE offers a cut-down version of the Desktop PC Internet Explorer,
- Symbian offer their own Web browser for EPOC OS,
- Java OS has Sun's own HotJava browser, although they will be offering a package of personal applications [38] to suit PersonalJava environments.

Figure 9 shows an example Web page (Bloomberg Financial News) on a desktop device using Microsoft Internet Explorer and on a 3Com Palm III device, using the ProxiNet browser.

When comparing Web browsers, various aspects determine their suitability for the intended applications. This may include some of the following:

- support for forms and frames,
- ability to run Java Applets with the required security level,
- support for image types, e.g. JPEG, GIF, Animated GIF,
- support for authentication and/or signing,
- protocols (and versions) supported (e.g. HTTP, FTP),
- additional facilities offered, e.g. option to zoom images on small devices.

On devices with limited user interface functions, the key question is the Web browsing capability provided. For example, a Palm III device has a display of 160 × 160 pixels with four possible grey levels. It provides a limited form of image output, but most images will not display well without transformation [39].

For wireless devices, standards for Web access are being developed by the Wireless Access Protocol (WAP) Forum [40]. WAP-compliant devices use a specialised wireless mark-up language (WML), with specially developed content or via a proxy translation from HTML to WML. They use a menu-driven user interface (micro-browser) to access Web text data (Fig 10).

Wireless Knowledge [41], a Microsoft and Qualcomm joint venture, has proposed a proprietary solution. All wireless-enabled devices connect to a central operations server, which provides interfaces to different networks (particularly attractive in the USA with a multiplicity of network operators and technologies) and device types. Qualcomm have agreed to work towards engineering Windows CE into their wireless handsets, to be sold under the Qualcomm brand in the USA and other brand names in foreign markets.

There are other approaches to delivering Web content data over wireless links. The Palm VII device (Fig 11), provides wireless Web access, via a proxy server, using a Web formatting technique called 'Web clipping' [42]. It is due to be released in 1999, initially in the USA, and a European service is expected later.



Fig 9 Bloomberg Financial News on desktop Web browser and Palm III ProxiNet browser.

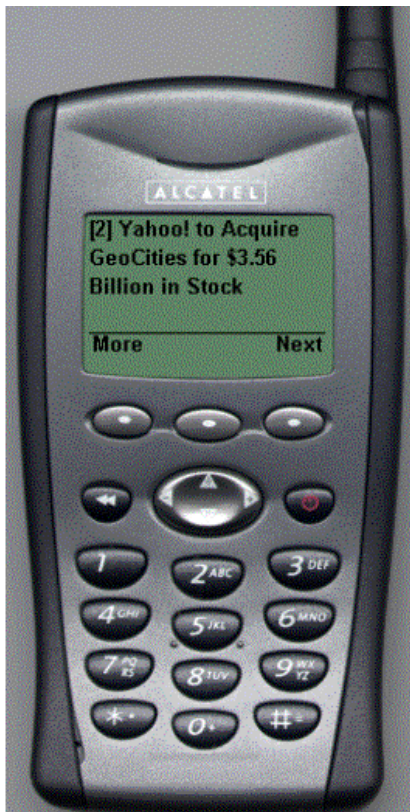


Fig 10 Bloomberg Financial News Web Page on WAP-enabled device.



Fig 11 Prototype Palm VII.

Symbian are aiming the EPOC OS at the wireless devices, with micro-browsers from STNC [43].

ETSI produced two categories of GSM Web access for the mobile station application execution environment (MExE) [44]. The first is based on the wireless application protocol, and the second on the PersonalJava environment, where more processing and user interface capabilities are available.

Web access options are becoming very diverse. If an eCommerce service is dependent on certain features, then either users need the correct Web-browsing facility or the

service has to adapt itself to the specific characteristics of the access device. This is the issue that adds significant complexity to eCommerce service delivery. Even though Web access may soon be available from many devices, it is essential that the eCommerce service remains presentable on the access device being used, either through adapting the content at the service provider end or within the device. Some possible techniques for achieving this will be discussed in section 3.

eCommerce devices

As device costs reduce, it is likely that enterprises, such as retailers and banks, will produce special-purpose eCommerce devices specifically for use with their services. As an example, Safeway Superstore in Basingstoke has started an electronic shopping trial called Easi-Order. Two hundred shoppers will be issued with Symbol PDAs, that are commercial versions of the 3Com Palm PDA, with the addition of a barcode scanner in the hood (Fig 12).



Fig 12 Safeway Easi-Order terminal.

A customised store download is available to each shopper's PDA, based on individual shopping preference. New items can be chosen and scanned in for immediate purchase, and are added to their profile to appear in their next downloaded list. This application has been designed to also run on other platforms including the EPOC OS.

3. Delivery techniques

3.1 Architectures

Delivering eCommerce services to a range of different capability access devices will require the use of a flexible, high-performance middleware architecture, capable of supporting the expanding range of network standards, device capabilities and user expectations. Key issues for eCommerce services are service provision and management, including updates and charging.

Some of the emerging architectures designed to support eCommerce services are described briefly below. All of these attempt to address the issue of access device diversity.

- Telecommunications Information Networking Architecture Consortium (TINA-C)

TINA [45] has a service architecture that is primarily designed to support telecommunications and information management services. The consortium is made up of telecommunications operators, equipment manufacturers, computer manufacturers and software companies. TINA access devices (computation objects) interact with other objects via the Distributed Processing Environment (DPE), which provides transparency of location, distribution and replication. Implementations of the DPE are strongly based on the common object request broker architecture (CORBA) protocol. TINA includes session and subscription models to support service provision. There have been a number of trial TINA platforms developed, mainly for providing telecommunications carrier services, but also including an electronic shopping mall, video on demand, and customer network management services.

- Jini

Jini [46] is Sun's recently announced network architecture. Jini is a distributed system that behaves as a network federation. A computing resource (e.g. disk storage, application) can register itself as a service to any other registered device within the federation. It provides look-up services, a service-leasing model, transactions and event models. It is based on Java technology, with Java Remote Method Invocation (RMI) as the inter-object protocol.

Java-based access devices integrate straightforwardly with the architecture, whereas non-Java based applications require a proxy. Several scenarios for the use of Jini have been put forward, most notably around the idea of renting applications [47].

- Wireless Knowledge

Wireless Knowledge [41] is a joint venture between Microsoft and Qualcomm. As described in section 2.3, its main focus is on ubiquitous wireless device access to central services. Its technology is based on the Microsoft BackOffice family and Microsoft Commercial Internet System (MCIS) and best suits devices that are based on a Microsoft OS (e.g. Microsoft Windows 98, Microsoft Windows CE). Wireless Knowledge's strength is in its working relationships with most wireless carriers in the US and covers all digital wide area network technologies, including GSM. The first architecture product, called Revolv, is expected to become available to commercial end-users in early 1999. It is unclear when this will become available in Europe. Microsoft has also described a home networking architecture called

'Universal Plug and Play' [48], which is based around their Windows 2000 platform. It is intended to be used to interconnect 'smart objects' around the home.

- IBM Transcode

IBM Transcode [49] is an implementation of IBM's idea of pervasive computing [2]. The middleware takes content that is written in a generic language, e.g. HTML, and transcodes it into a form suitable for the access device. For this it uses a proxy server, which is also able to complete other operations on behalf of the client, for example, providing continued connection over an intermittent network link.

3.2 Access device capability identification

In order to adapt themselves for a range of devices, services need to be able to access some form of device profile, which defines the capabilities of a specific device. Work on access device profiles is being carried out by a number of groups, including W3C [50] and MNCRS [51].

W3C proposes an XML/RDF encoding of the device capabilities which can either originate from the device or be found on a network server. These capabilities include user preferences (e.g. preferred language), hardware (e.g. screen size) and software (e.g. level of HTML support). The MNCRS is adopting a different Java-oriented approach, by providing an API, enabling applications, proxies and servers to interrogate devices directly.

There are only trial implementations of these specifications as yet, but agreement and adoption of standards in this area is a key requirement for flexible eCommerce service delivery.

3.3 Adaptation

Current Internet services often assume the use of desktop PCs, with large screen, CPU, memory and reasonable bandwidth networks. With lower capability devices and networks the services need to be adapted to work on them, and various approaches to service adaptation are being developed. Because device capabilities will continue to change as new devices are introduced, there needs to be some form of generic service adaptation capability to automatically discover and adapt the service to the device capabilities.

Reflection

One proposed approach to automatic adaptation is through the use of a software technique known as 'reflection', first developed by Smith [52]. Reflection refers to the ability to examine and manipulate the state of an executing program. The execution state is represented as data that can be inspected, and a change made to that data is

reflected back as a corresponding change in the system behaviour. Through the use of reflection middleware processes can be inspected and manipulated, with the objective of achieving the required end-to-end system performance.

When applied to object-based systems, reflection requires the use of a meta-level interface, to open up and interpret the underlying object (Fig 13).

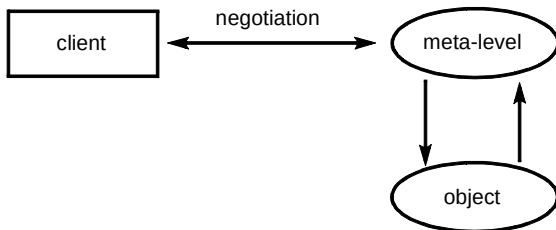


Fig 13 Meta-level interface to an object.

For example, the client negotiation with the meta-level could involve a request for the client device screen size or memory capability. This can take place in two ways:

- an additional interface is added to the object, called the meta-object protocol (MOP), through which the negotiation takes place — the object that is then downloaded to the device will already have been adapted and will immediately be available to be executed,
- the complete object is downloaded for the adaptation to take place on the client device.

There is a trade-off between incurring additional network traffic on a potentially low-bandwidth link, or additional processing on a potentially lower processor capable device. There is also an additional MOP interface required with the first option, but this would give a more flexible interface, as opposed to the predefined adaptation with the second option.

The second option is sometimes referred to as introspection. Firstly, the object exists with adaptation capability developed within it. Then, when the object executes, it reflects on its own representation and the run-time environment, and, using predetermined rules, changes its representation (Fig 14). In the case of limited capability devices, it would adapt itself based on parameters such as screen size and available memory.

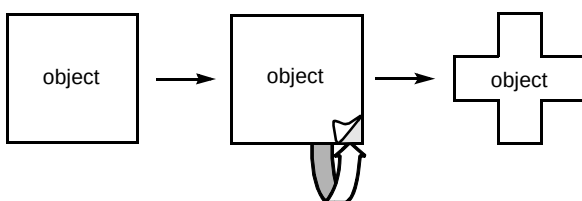


Fig 14 Introspection process.

An experimental middleware platform, FlexiNet [53, 54], developed by the ANSA consortium, has reflective and introspective capabilities that enable automatic service adaptation to be implemented.

4. Conclusions

It is clear from the discussion presented in this paper that there is significant diversity in the range of devices likely to be used to access eCommerce services. As information appliances and other personal computing devices become part of everyday life there will be an expectation that eCommerce services can be accessed using them, whatever their form; eCommerce services are currently being produced in expedient ways, but future services and devices will need to satisfy the full eCommerce requirements, such as security, communication flexibility, charging and service interoperation.

Many public eCommerce access devices such as kiosks and ATMs are intended primarily for eCommerce, and offer few other services. Many personal devices, such as desktop PCs, will continue to be multi-purpose, but it is likely that smaller and cheaper devices may become marketable as specific eCommerce access devices, such as the Safeway Easi-Order device described above.

In the future it will also make sense to use different access devices to support each other, for example, using Web information accessed from a personal device to locate the nearest ATM [55]. This type of application demonstrates the value of building eCommerce applications that co-operate. The range of devices that may be used to access eCommerce services needs to be understood, so that the service can be delivered in the most convenient way.

This paper has considered the range of devices that are likely to be used to access eCommerce services in the near future, and some of the issues that will arise for service providers as a result. However, this proliferation of devices will present significant opportunities in providing eCommerce services that meet the needs of consumers in more convenient and innovative ways. The key will be in producing reliable and effective services, to capture consumer interest and build confidence, using information appliances that meet their needs.

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